Traceability Supplement

TABLE XIV
PROPOSED SOURCE-DERIVED TRACEABILITY DIMENSIONS (UNMEASURED)

ID	Required correspondence	Unit numerator / denominator
TR01	identity/activity binding to task, exact artifact revision/digest, activity, and actor	complete slots / required slots
TR02	request/input → rule version → evaluated check/result → decision	complete chains / inventoried decisions
TR03	decision → observed state or artifact effect with before/after evidence	complete correspondences / decisions requiring outcome
TR04	triggering failure → corrective package → exact revalidation → result	complete repair chains / actual repair attempts
TR05	forward requirement → artifact → verification and reverse in-scope change mapping	report forward and reverse separately
TR06	non-author independent exercise of declared audit/replay target	exercised targets / declared targets

Proposed and unmeasured; codebooks/source rationale: Data and Code Availability. No score was collected or validated.

I. PROPOSED TRACEABILITY INSTRUMENT (NOT MEASURED)

Traceability tests recover requirement/artifact/action/outcome correspondence, not correctness, hidden reasoning, developer skill, or player benefit. For each dimension d, freeze the arm-neutral observer inventory U_d and required links $L_d(u)$ for each unit u. Let N_d count applicable units in U_d and n_d count those whose required links are all verified. Independent assessors award 1 only for exact source/version, all raw-evidence relations, and locator; otherwise 0 with missing/inaccessible/stale/contradicted/unverified reasons. Report n_d/N_d only for a complete inventory and $N_d > 0$, alongside disagreement and seeded-corruption controls; no weighted composites. An incomplete inventory is INDETERMINATE, with observed counts and the inventory gap reported. N/A requires independent evidence of non-applicability; for repairs, independently verified zero actual repair attempts qualifies, but an empty log does not. Distinguish PASS, FAIL, NOT_RUN, STAGE_NOT_REACHED, OUT_OF_SCOPE, and MISSING; `stage_not_reached` never passes.